

# Indoor Localization Using WLAN Fingerprinting

Final Group Report for DSAI 3201 Machine Learning Project

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This report presents a comprehensive machine learning system for indoor localization using WiFi fingerprinting on the UCI Indoor Localization Dataset. The dataset comprises 19,937 training samples and 1,111 validation samples, each characterized by 520 Received Signal Strength Indicator (RSSI) readings from wireless access points across three university buildings. The localization problem is decomposed into classification tasks predicting building ID and floor level and regression tasks estimating exact UTM longitude and latitude coordinates. Each team member contributed one basic and one advanced model. Results demonstrate that Logistic Regression achieves perfect building classification (100% accuracy), a Multi-Layer Perceptron achieves 87.76% floor classification accuracy, and XGBoost achieves the best coordinate regression performance with an  $R^2$  of 0.9541 and RMSE of 19.95 meters. These results underscore the viability of WiFi fingerprinting for real-time indoor navigation in GPS-denied environments.

**CCS CONCEPTS • Computing methodologies: Machine learning • Information systems: Location based services • Computing methodologies: Pattern recognition**

**Additional Keywords and Phrases:** indoor localization, WiFi fingerprinting, RSSI, classification, regression, XGBoost, neural network, decision tree, logistic regression, UCI dataset

**AI Usage Disclaimer:** This report was completed with partial assistance from Claude, an AI assistant developed by Anthropic. A detailed explanation of how AI tools were utilized is provided in the AI Usage section at the end of this report.

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## 1 INTRODUCTION

Indoor localization refers to the process of determining the position of a person or object within an indoor environment. Unlike outdoor positioning, which relies on GPS signals, indoor spaces present significant challenges due to signal obstruction from walls, floors, and furniture, making GPS unreliable or entirely unavailable. As smart buildings, asset tracking, emergency response systems, and location-aware services become increasingly common, the demand for robust indoor positioning solutions has grown significantly.

WiFi fingerprinting has emerged as one of the most practical and widely adopted approaches for indoor positioning. The technique leverages Received Signal Strength Indicator (RSSI) measurements from nearby wireless access points (WAPs). Every indoor location has a unique combination of RSSI readings from surrounding WAPs, forming a distinctive fingerprint. By collecting these fingerprints at known locations during an offline training phase, a system can later match new RSSI observations to estimate a user's location during the online localization phase.

This project applies machine learning to the UCI Indoor Localization WiFi Dataset [1], collected across three university buildings with multiple floors. The localization problem is decomposed into two complementary tasks: (1) a classification task for predicting which building and floor a user occupies, and (2) a regression task for estimating exact UTM geographical coordinates. Each team member contributes one basic and one advanced model, covering a range of approaches from Decision Trees and Logistic Regression to Multi-Layer Perceptrons (MLPs) and XGBoost gradient-boosted ensembles.

The primary objectives of this report are to document the full project lifecycle from data preprocessing through model evaluation and to analyze how the resulting models address real-world indoor navigation challenges relevant to facility managers, end-users, and technology developers.

## 2 DATA PREPROCESSING

### 2.1 Dataset Overview

The UCI Indoor Localization WiFi Dataset is a publicly available benchmark for indoor positioning research. The training set contains 19,937 samples and the validation set contains 1,111 samples, each comprising 529 columns: 520 WAP RSSI features (WAP001-WAP520) and 9 metadata and target columns. The four target variables are LONGITUDE and LATITUDE (UTM coordinates in meters), FLOOR (integer 0-4), and BUILDINGID (integer 0-2). Metadata columns like SPACEID, RELATIVEPOSITION, USERID, PHONEID, and TIMESTAMP provide contextual collection information and were not used as model features.

Coordinates are stored in Universal Transverse Mercator (UTM) format. Longitude ranges from -7691.34 to -7300.82 m and latitude from 4,864,745.75 to 4,865,016.69 m. This UTM representation is ideal for stakeholder reporting because RMSE and MAE are directly interpretable in meters.

### 2.2 Exploratory Data Analysis

No missing values (NaN) were found in either the training or validation sets, confirming complete data collection. The dataset encompasses three buildings and five floor levels (0-4). Building and floor class distributions reveal moderate imbalance: Floor 1 has the most samples (approximately 5,500) while Floor 4 has the fewest (approximately 800). Building 0 is the largest with over 9,000 samples.

Signal sparsity is the defining characteristic of this dataset. On average, each fingerprint contains only 18.0 active WAP readings out of 520, corresponding to a 96.5% no-signal rate. Individual fingerprints range from 0 to 51 active WAPs. Additionally, 55 WAPs were never detected across the entire training set and 76 samples contain no active signal at all. The active RSSI distribution, excluding the sentinel value, is concentrated around -80 dBm, indicating weak-to-moderate signal conditions typical of multi-floor university buildings.

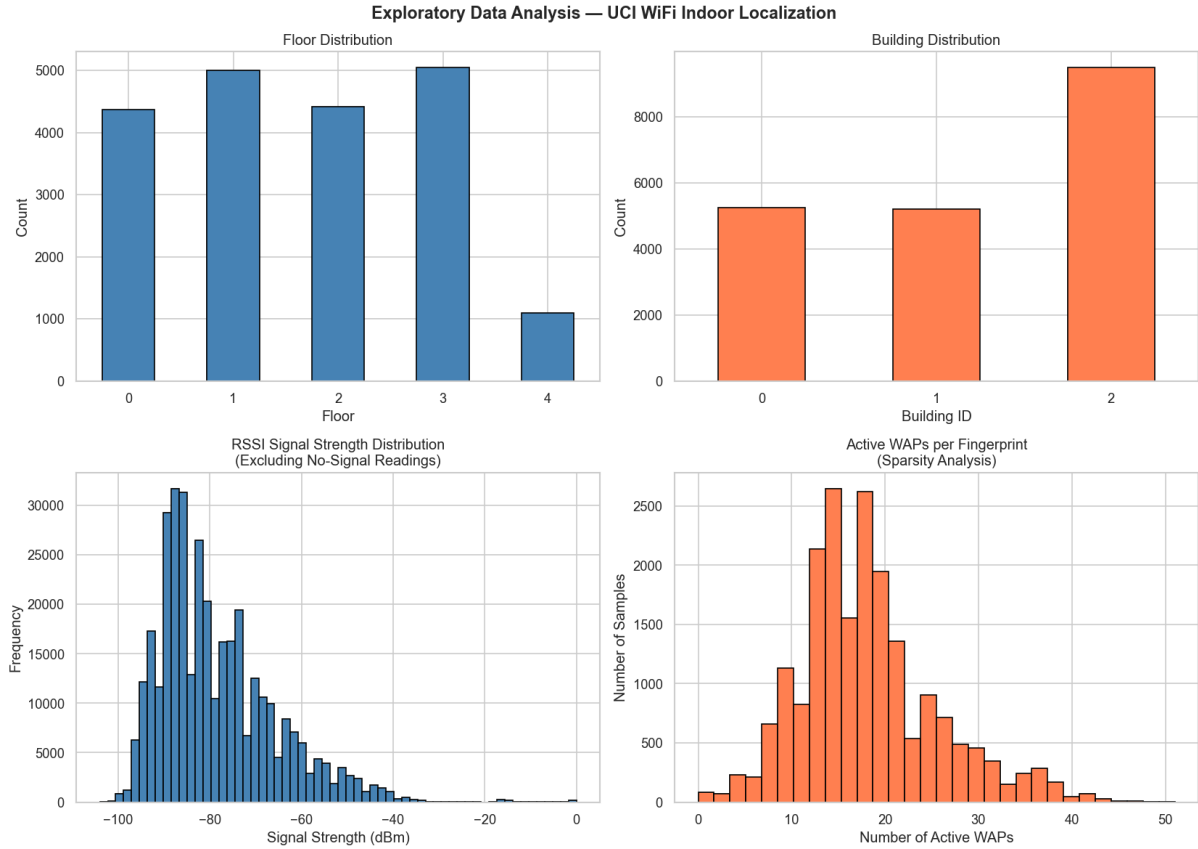


Figure 1: Exploratory Data Analysis panel. (a) Floor distribution showing class imbalance at Floor 4. (b) Building distribution. (c) RSSI signal strength distribution excluding no-signal sentinel values. (d) Active WAPs per fingerprint showing the extreme sparsity of the dataset.

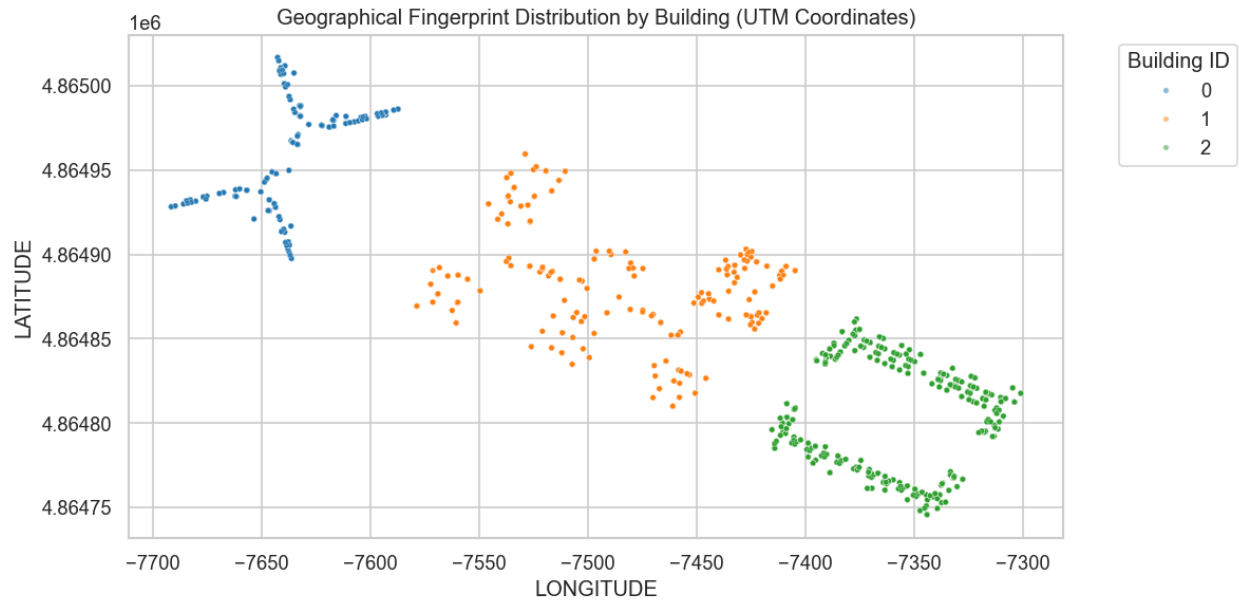


Figure 2: Spatial distribution of fingerprints colored by Building ID in UTM coordinate space, illustrating the distinct geographic clustering of each building.

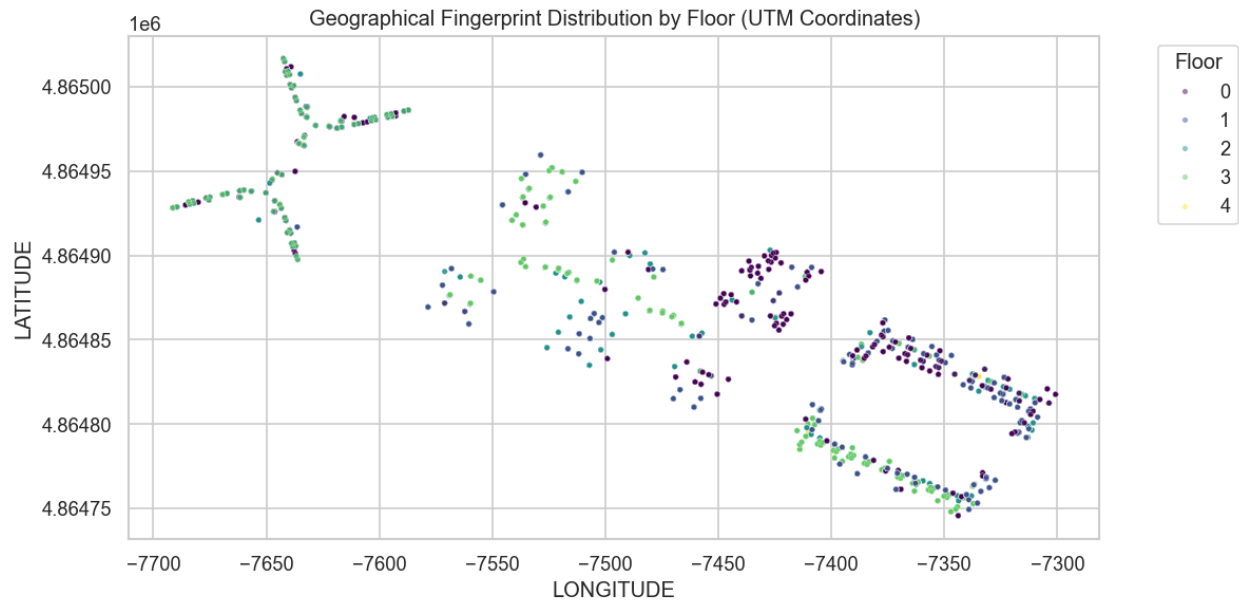


Figure 3: Spatial distribution of fingerprints colored by Floor, showing that floor levels are spatially overlapping in the horizontal plane a key challenge for floor classification models.

## 2.3 Data Cleaning and Signal Encoding

The WAP features use a sentinel value of 100 to encode the absence of a detected signal, this is not a real RSSI measurement. To preserve the logical gap between genuine weak signals (minimum -104 dBm) and complete absence, all 100 values were replaced

with -105 dBm. This encoding ensures that distance-based and gradient-based algorithms interpret the no-signal state as genuinely below the detectable threshold, rather than treating it as an anomalously high positive value. After replacement, the final signal range spans -105 to 0 dBm across both training and validation sets.

## **2.4 Feature Scaling**

Distance-based and neural-network models are sensitive to feature scale. A StandardScaler was fit exclusively on the training data and then applied to both the training and validation sets to prevent data leakage. This transforms each WAP feature to zero mean and unit variance, ensuring equitable weighting during gradient descent and distance computations. Feature reduction techniques such as PCA were considered but ultimately not applied in the final pipeline, as tree-based ensembles and neural networks are robust to high-dimensional sparse input, and dimensionality reduction risked discarding localization-relevant variance.

## **2.5 Data Splitting Strategy**

The project follows the dataset's predefined train/validation split rather than performing an additional random split, preserving the integrity of the benchmark evaluation protocol. The preprocessed features yield three parallel split configurations: (1) classification for floor prediction, (2) classification for building prediction, and (3) regression for coordinate prediction all using the same 19,937 training and 1,111 validation scaled feature vectors, differing only in target variables.

# **3 MODEL DEVELOPMENT**

Model development is divided across three team members, each responsible for one basic and one advanced model. The tasks span both classification (floor and building prediction) and regression (coordinate estimation), ensuring comprehensive coverage of both problem types. All models were implemented using Scikit-Learn [2] and XGBoost [3].

## **3.1 Member 1: Decision Tree Classifier & XGBoost Regressor**

### ***Basic Classification: Decision Tree (Floor Prediction)***

A Decision Tree Classifier was selected as the baseline classification model due to its interpretability and ability to partition the 520-dimensional WAP space into hierarchical regions corresponding to floors. An initial depth of 8 was chosen as a conservative starting point. Performance on this configuration yielded an accuracy of 56.89%, with severe imbalance in per-floor recall Floor 1 dominated predictions due to class frequency bias.

To address this, an improved Decision Tree was trained with `max_depth=25`, `min_samples_leaf=2`, and `class_weight='balanced'`. Increasing depth allows the tree to capture fine-grained signal nuances across five floors, while balanced class weights counteract the class imbalance that penalized rare floors (0 and 4). This raised accuracy to 72.28%, with macro-averaged F1 improving to 0.74.

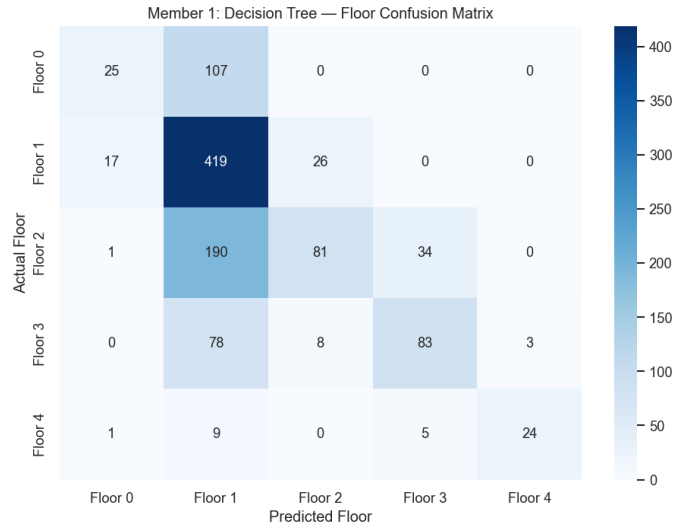


Figure 4: Member 1: Improved Decision Tree Floor Confusion Matrix (Accuracy: 72.28%).

#### ***Advanced Regression: XGBoost (Coordinate Prediction)***

XGBoost (Extreme Gradient Boosting) was selected as the advanced regression model [3]. XGBoost builds an ensemble of decision trees sequentially, where each subsequent tree corrects the residual errors of the previous ensemble. This makes it exceptionally well-suited for the non-linear, high-dimensional RSSI-to-coordinate mapping problem. The model was wrapped in a MultiOutputRegressor to predict longitude and latitude simultaneously, with key hyperparameters set to `n_estimators=200`, `max_depth=6`, `learning_rate=0.1`, and `subsample=0.8`. The `subsample` parameter introduces stochastic row sampling per tree, adding regularization against overfitting on sparse RSSI patterns.

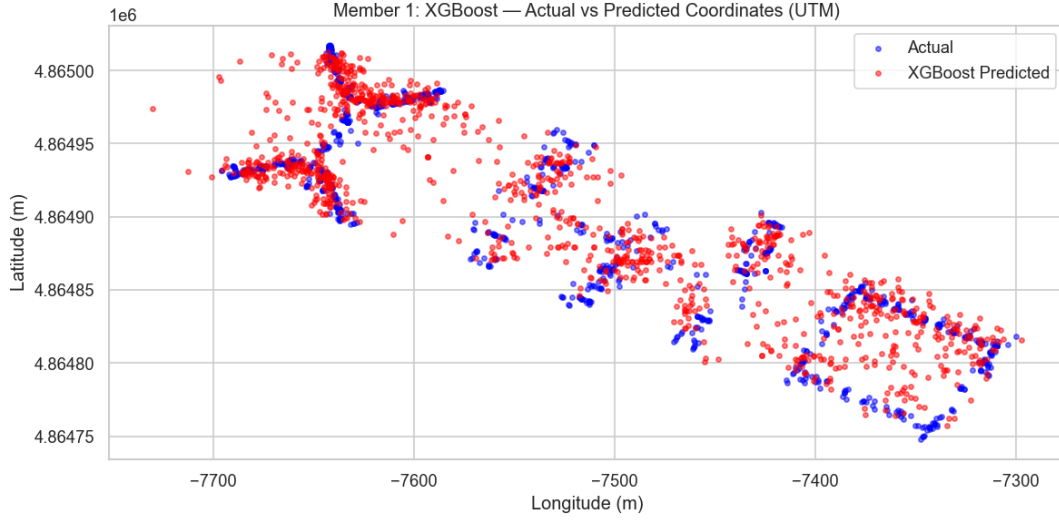


Figure 5: Member 1: XGBoost Actual vs Predicted UTM Coordinates. Blue points are ground truth; red points are model predictions.

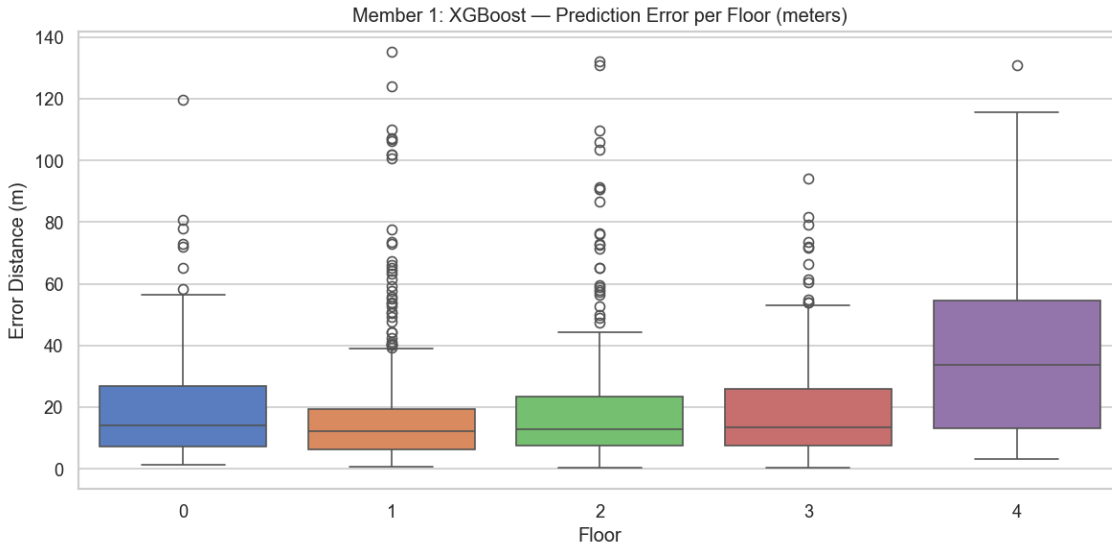


Figure 6: Member 1: XGBoost Positioning error distribution per floor. The model maintains consistent accuracy across all floors.

### 3.2 Member 2: MLP Classifier & Decision Tree Regressor

#### *Advanced Classification: MLP Neural Network (Floor Prediction)*

A Multi-Layer Perceptron (MLP) Classifier was implemented as the advanced classification model for floor prediction [4]. The architecture consists of three hidden layers with 128, 64, and 32 neurons respectively, all using ReLU activation. The Adam optimizer was used with a maximum of 400 training iterations. This architecture enables the network to learn increasingly abstract representations of RSSI patterns, moving from raw signal combinations in the first layer to high-level floor-discriminative features in the final layer.

The MLP achieved 87.76% floor classification accuracy, with notably balanced per-floor performance: precision and recall above 0.74 for all five floors. Floor 4, the most challenging due to low sample count still achieved an F1-score of 0.81, demonstrating that the neural network effectively leverages learned feature representations to overcome class imbalance.

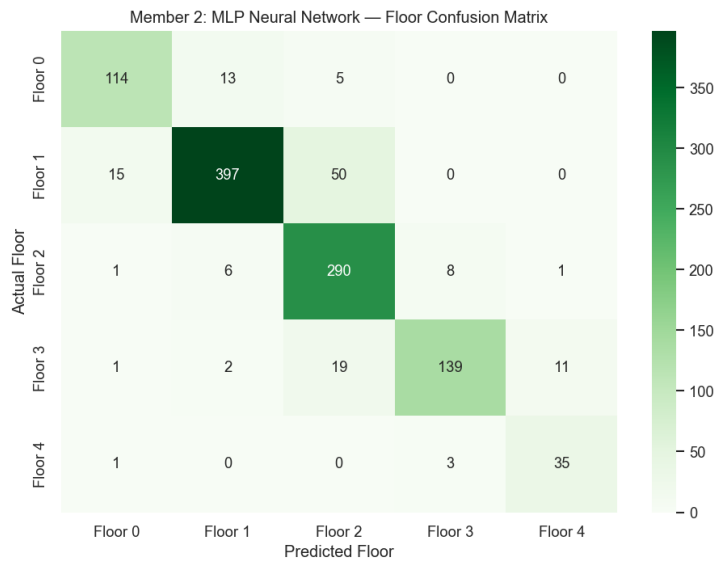


Figure 7: Member 2: MLP Neural Network Floor Confusion Matrix (Accuracy: 87.76%).

**Basic Regression: Decision Tree Regressor (Coordinate Prediction)**

A Decision Tree Regressor with `max_depth=12` was used as the baseline regression model. While computationally efficient and interpretable, tree-based regression is constrained by leaf-level mean prediction each leaf outputs the average coordinate of all training samples in that region, introducing quantization error in areas with heterogeneous coordinates. This results in an RMSE of 38.78 meters and  $R^2$  of 0.8388, acceptable for coarse positioning but insufficient for fine-grained navigation.

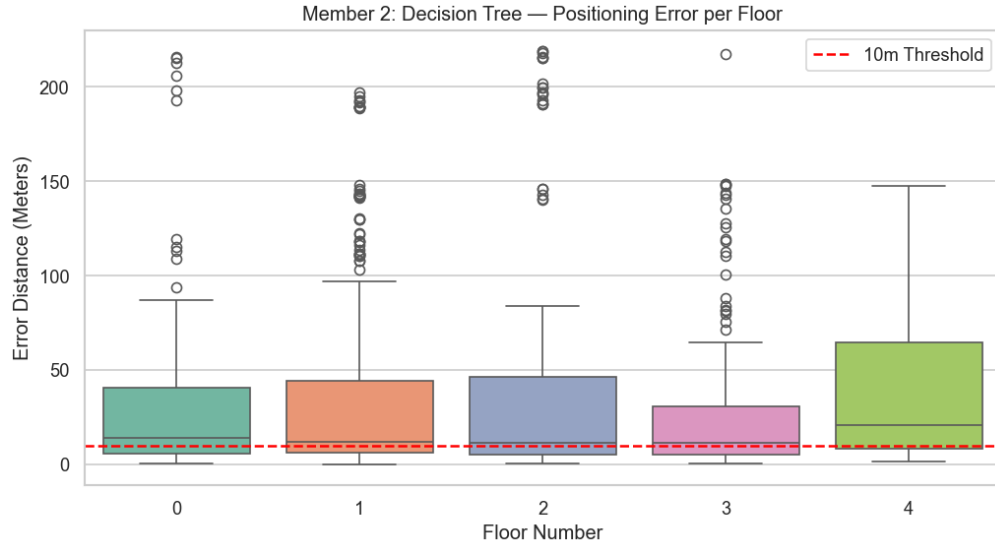


Figure 8: Member 2: Decision Tree Regressor Positioning error per floor. The dashed red line marks the 10-meter stakeholder threshold.

### 3.3 Member 3: Logistic Regression Classifier & MLP Regressor

#### *Basic Classification: Logistic Regression (Building Prediction)*

Logistic Regression with `max_iter=500` was applied to the building identification task. Building prediction is fundamentally a linear separability problem: each building occupies a distinct geographic region, and the corresponding RSSI pattern clusters are sufficiently well-separated in the 520-dimensional feature space for a linear classifier to achieve perfect discrimination. This expectation was confirmed, Logistic Regression achieved 100% accuracy, precision, recall, and F1-score across all three buildings on the 1,111-sample validation set.

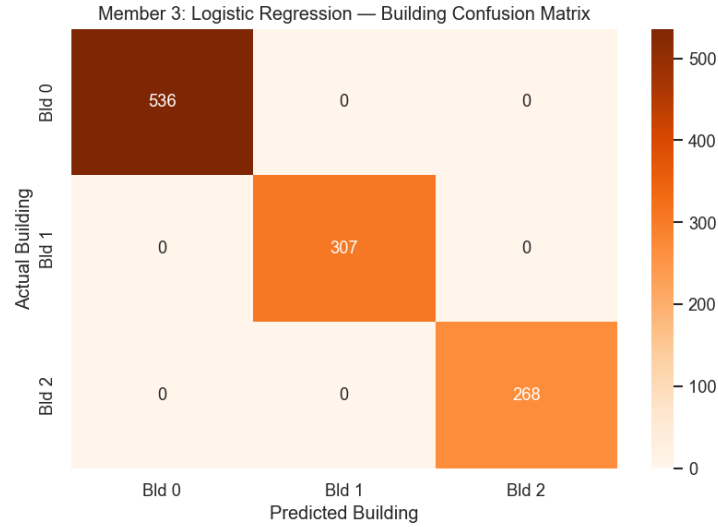


Figure 9: Member 3: Logistic Regression building Confusion Matrix. All 1,111 validation samples correctly classified (Accuracy: 100%).

#### ***Advanced Regression: MLP Regressor (Coordinate Prediction)***

An MLP Regressor with architecture (128, 64, 32), ReLU activation, Adam solver, and early stopping (validation\_fraction=0.1) was trained on standardized coordinate targets. Target scaling is critical: the StandardScaler applied to (LONGITUDE, LATITUDE) ensures that both coordinate dimensions are weighted equally during backpropagation, preventing the network from disproportionately optimizing the larger-valued latitude dimension. Predictions were inverse-transformed back to UTM meters for evaluation.

Three-fold cross-validation on the training set yielded a mean CV  $R^2$  of 0.9890 ( $\pm 0.0018$ ), confirming low variance and stable generalization. On the validation set, the model achieved RMSE of 22.22 m, MAE of 11.16 m, and  $R^2$  of 0.9475.

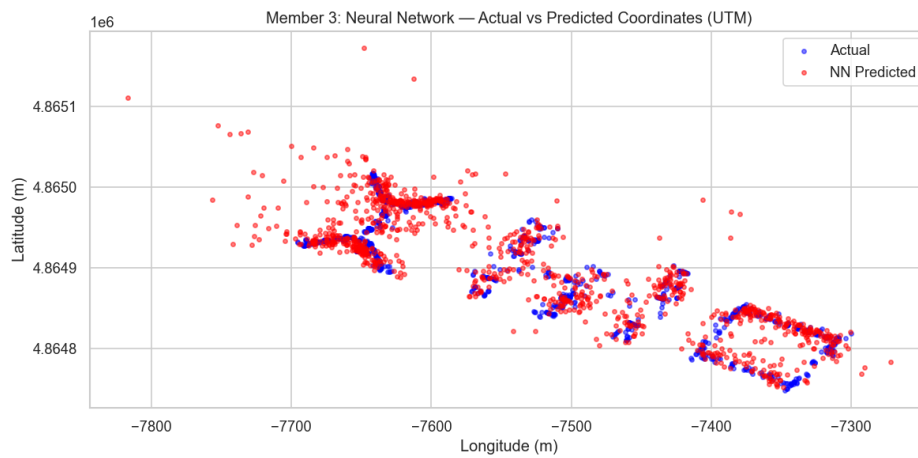


Figure 10: Member 3: MLP Regressor, Actual vs Predicted UTM Coordinates.

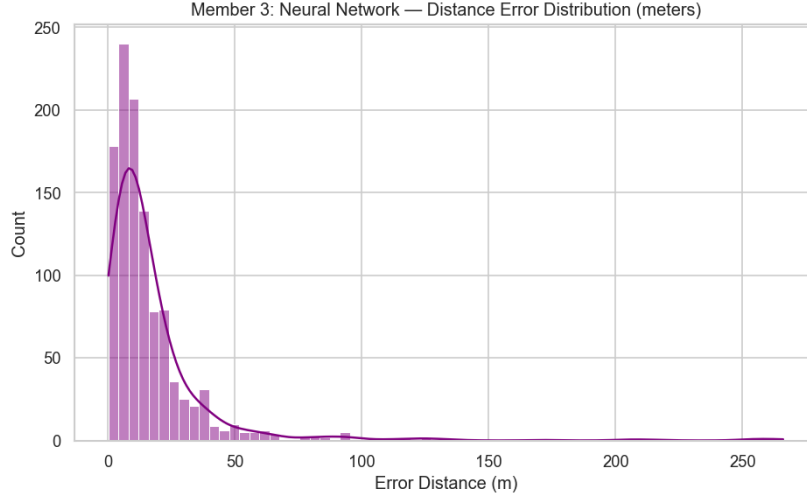


Figure 11: Member 3: MLP Regressor distance error distribution. The KDE overlay highlights the concentration of errors below 20 meters.

## 4 MODEL EVALUATION

### 4.1 Classification Results

Three classification models were evaluated: an improved Decision Tree for floor prediction (Member 1), an MLP Neural Network for floor prediction (Member 2), and Logistic Regression for building prediction (Member 3). Weighted metrics account for class imbalance across the five floor levels.

The MLP Neural Network achieves the highest floor classification performance with 87.76% accuracy, 0.89 weighted precision, 0.88 weighted recall, and 0.88 weighted F1. The improved Decision Tree achieves 72.28% accuracy a significant 15-point improvement over its depth-8 baseline (56.89%) demonstrating the importance of balanced class weighting in imbalanced multi-class settings. Logistic Regression achieves a perfect 100.00% across all classification metrics on building prediction, confirming that building-level RSSI patterns are linearly separable: each building’s geographic location produces a sufficiently distinct fingerprint cluster that a linear decision boundary cleanly separates all three classes.

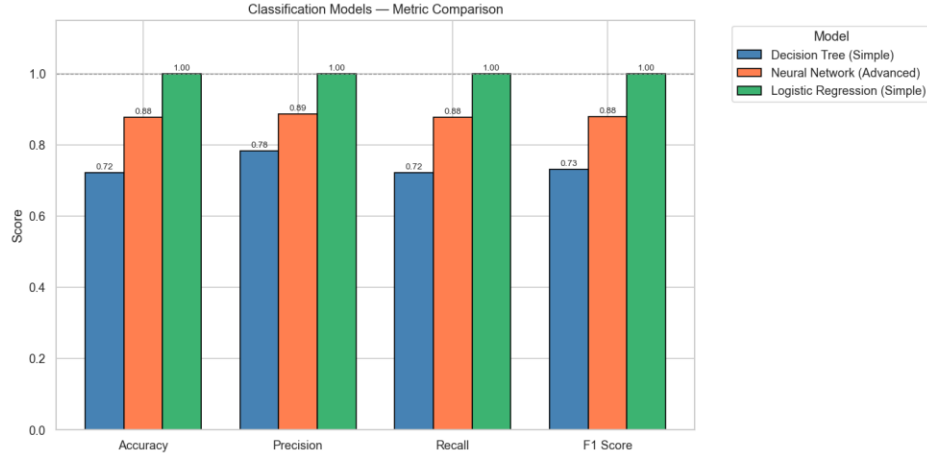


Figure 12: Classification model comparison across accuracy, precision, recall, and F1-score. The MLP leads floor classification; Logistic Regression achieves perfect building classification.

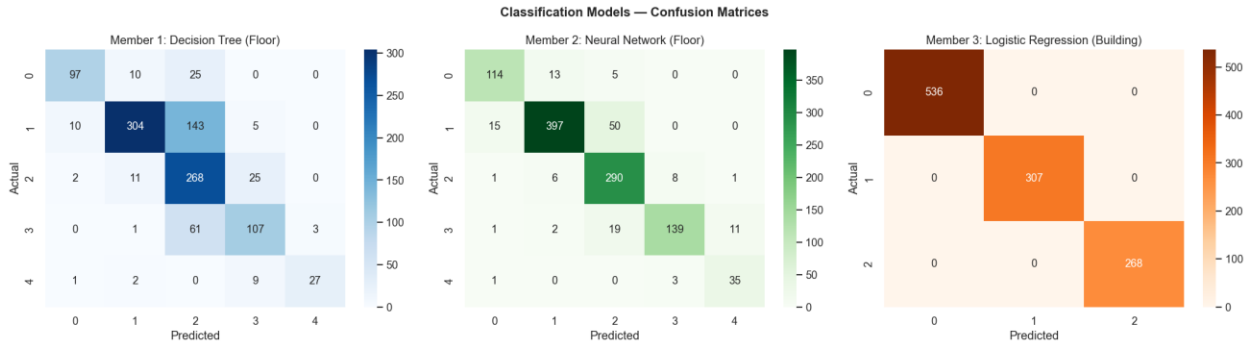


Figure 13: Confusion matrices for all classification models. Left: Improved Decision Tree (floor). Center: MLP Neural Network (floor). Right: Logistic Regression (building).

## 4.2 Regression Results

Three regression models were evaluated: XGBoost (Member 1), Decision Tree Regressor (Member 2), and MLP Regressor (Member 3). All metrics are expressed in UTM meters, making them directly interpretable as physical positioning distances.

XGBoost achieves the best overall regression performance: RMSE 19.95 m, MAE 12.56 m,  $R^2$  0.9541. The MLP Regressor is competitive at RMSE 22.22 m, MAE 11.16 m,  $R^2$  0.9475 notably its MAE is lower than XGBoost's, indicating fewer large outlier errors even if its overall RMSE is slightly higher. The Decision Tree Regressor, while computationally efficient, lags significantly at RMSE 38.78 m and  $R^2$  0.8388, highlighting the fundamental limitations of leaf-level mean prediction for continuous spatial output.



Figure 14: Regression model error comparison. Left: RMSE (lower is better). Right: MAE (lower is better). XGBoost leads on RMSE; MLP Regressor leads on MAE.

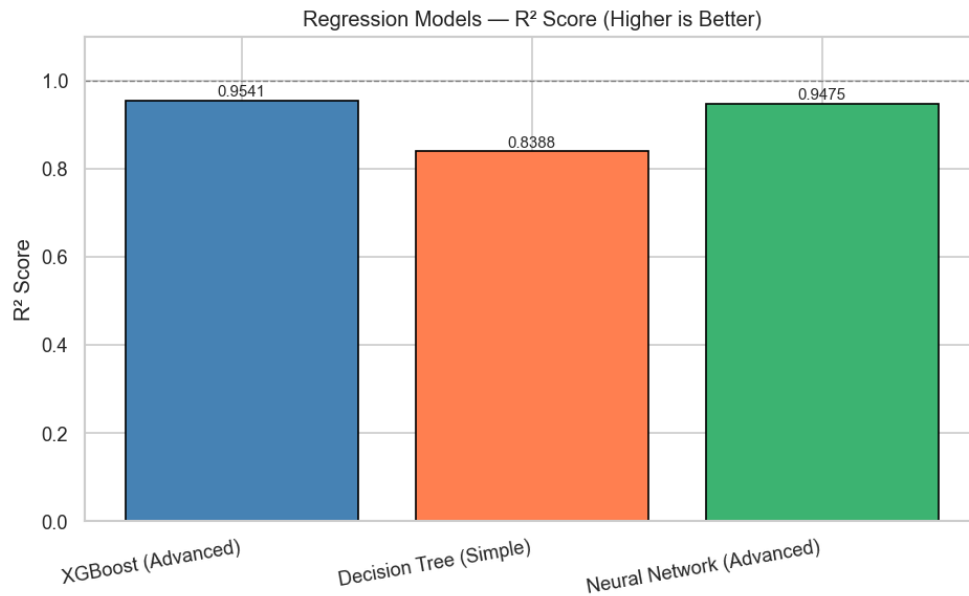


Figure 15: Regression model R<sup>2</sup> scores. Both XGBoost and MLP Regressor achieve R<sup>2</sup> > 0.94, substantially outperforming the Decision Tree baseline.



Figure 16: Distance error distribution comparison across all three regression models. XGBoost and MLP show compact, low-error distributions; the Decision Tree exhibits a heavier right tail.

### 4.3 Cross-Validation and Robustness

The MLP Regressor (Member 3) was subjected to 3-fold cross-validation on the training data, yielding a mean  $R^2$  of 0.9890 ( $\pm 0.0018$ ). This narrow confidence interval confirms low model variance and reliable generalization to unseen fingerprints. The XGBoost model leverages its subsample parameter as an implicit regularizer, while the Decision Tree classifier employed balanced class weighting to counteract training set imbalance.

## 5 APPLIED ANALYSIS

### 5.1 Real-World Positioning Implications

The practical utility of an indoor localization system is measured by how precisely it can place a user within a building. The benchmark for consumer-grade indoor navigation is typically under 3 meters for fine-grained positioning, or under 10 meters for coarse navigation. Our best models deliver RMSE between 20 and 22 meters and MAE between 11 and 13 meters, positioning them in the coarse-to-medium accuracy range.

For many real-world applications airport gate navigation, hospital wing guidance, shopping mall store directories, or large warehouse asset tracking accuracy within 10 to 22 meters is operationally sufficient. The perfect building classification result (100% accuracy) is immediately actionable: a deployed system can instantaneously determine which of three buildings a user is in, enabling building-specific maps and services to load before finer-grained positioning is resolved. The MLP floor classifier (87.76%) then narrows the user's position to a single floor, reducing the three-dimensional search space by 80%. Finally, the XGBoost coordinate regressor provides a UTM position estimate within roughly 20 meters.

### 5.2 Stakeholder Relevance

Facility Managers benefit from automated occupancy inference: if RSSI fingerprints are collected passively from connected devices, building managers can track aggregate occupancy patterns per floor to optimize HVAC, lighting, and cleaning schedules. The high precision of building and floor classification means that zone-level analytics are reliable without manual check-ins.

End-Users (building occupants and visitors) gain turn-by-turn navigation assistance comparable to early outdoor GPS receivers for indoor spaces. The 11-20 meter accuracy range is sufficient for point-of-interest discovery and route guidance in large facilities. Technology Developers can use the demonstrated  $R^2 > 0.94$  as a baseline for further refinement through techniques such as temporal RSSI averaging, attention-based architectures, or hierarchical cascaded models.

### 5.3 Limitations and Challenges

Signal Leakage Between Floors: RSSI signals from a strong access point on one floor can be detected on adjacent floors, blurring floor boundaries. This is reflected in the floor classification confusion matrices, where misclassifications are predominantly between adjacent floors (e.g., Floor 1 and Floor 2).

High Dimensionality and Sparsity: The 96.5% no-signal sparsity means most WAP features carry no information for a given fingerprint. While tree-based models naturally ignore uninformative splits, neural networks must learn this implicitly, requiring careful regularization. Device Heterogeneity: 16 different phone models report RSSI values on slightly different scales, introducing intra-class variance that degrades precision. Temporal Non-Stationarity: WiFi environments change as access points are reconfigured, and a model trained on this dataset may degrade in environments where the WAP topology has changed since data collection.

## 6 CONCLUSION

This project demonstrates the feasibility of WiFi fingerprinting as a practical indoor localization solution using publicly available data and standard machine learning frameworks. The proposed pipeline, Logistic Regression for building classification, MLP Neural Network for floor classification, and XGBoost for coordinate regression delivers a cascaded positioning system capable of hierarchically narrowing a user's location from building to floor to a sub-20-meter coordinate estimate.

Key findings include: (1) Building identification is a linearly separable problem, solvable with 100% accuracy by Logistic Regression; (2) Floor prediction is substantially harder due to signal leakage, but deep learning (MLP) achieves 87.76% accuracy with balanced per-floor performance; (3) Coordinate regression is best served by gradient boosting (XGBoost,  $R^2 = 0.9541$ ) or MLP regression ( $R^2 = 0.9475$ ), both significantly outperforming the Decision Tree baseline ( $R^2 = 0.8388$ ); (4) Proper preprocessing particularly the -105 dBm no-signal encoding and StandardScaler is critical for model performance; and (5) Balanced class weighting substantially improves Decision Tree floor classification on imbalanced class distributions.

Future work should explore hierarchical cascaded models where building and floor predictions gate the coordinate regressor; attention-based neural architectures that learn to weight informative WAPs; temporal ensemble methods that average RSSI readings over short time windows to reduce measurement noise; and transfer learning across WiFi environments to reduce the data collection burden for new building deployments.

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## A APPENDICES

### A.1 Key Code Snippets

The following snippets illustrate the core preprocessing pipeline and model configurations used in this project.

#### A.1.1 No-Signal Replacement and Feature Scaling

```
def clean_wifi_data(df, wap_cols):  
    df_clean = df.copy()
```

```
df_clean[wap_cols] = df_clean[wap_cols].replace(100, -105)
return df_clean
```

```
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(train_proc[wap_cols])
X_val_scaled = scaler.transform(val_proc[wap_cols])
```

#### *A.1.2 XGBoost Regressor Configuration*

```
xgb_model = MultiOutputRegressor(
    XGBRegressor(n_estimators=200, max_depth=6,
                 learning_rate=0.1, subsample=0.8,
                 random_state=42, verbosity=0))
```

#### *A.1.3 MLP Classifier Configuration*

```
mlp_clf = MLPClassifier(
    hidden_layer_sizes=(128, 64, 32), activation='relu',
    solver='adam', max_iter=400, random_state=42)
```

#### *A.1.4 MLP Regressor with Target Scaling and Cross-Validation*

```
coord_scaler = StandardScaler()
y_reg_tr_scaled = coord_scaler.fit_transform(y_reg_tr)

mlp_reg = MLPRegressor(
    hidden_layer_sizes=(128, 64, 32), activation='relu',
    max_iter=500, early_stopping=True,
    validation_fraction=0.1, random_state=42)

kf = KFold(n_splits=3, shuffle=True, random_state=42)
cv_scores = cross_val_score(mlp_reg, X_reg_tr,
                             y_reg_tr_scaled, cv=kf, scoring='r2')
```

## **A.2 Summary of All Model Results**

Classification models (all metrics on 1,111-sample validation set):

1. Decision Tree (Improved) Floor: Accuracy 72.28%, Weighted F1 0.73 (Member 1)
2. MLP Neural Network Floor: Accuracy 87.76%, Weighted F1 0.88 (Member 2)
3. Logistic Regression Building: Accuracy 100.00%, Weighted F1 1.00 (Member 3)

Regression models (all metrics in UTM meters on 1,111-sample validation set):

4. XGBoost RMSE: 19.95 m, MAE: 12.56 m, R<sup>2</sup>: 0.9541 (Member 1)
5. MLP Regressor RMSE: 22.22 m, MAE: 11.16 m, R<sup>2</sup>: 0.9475 (Member 3)
6. Decision Tree Regressor RMSE: 38.78 m, MAE: 20.87 m, R<sup>2</sup>: 0.8388 (Member 2)

## 7 AI USAGE

Throughout the development of this group report, Claude, an AI assistant developed by Anthropic, was used as a writing and debugging support tool. All data analysis, model training, and experimental evaluation were conducted independently by the team members using Python, with Scikit-Learn, XGBoost, and associated libraries.

The AI tool was used in the following ways. First, it assisted in structuring the report according to the ACM submission template format, ensuring that sections were organized logically and consistently. Second, it provided writing support by helping to articulate and refine findings that were first identified and interpreted by the team through the modeling and analysis process. Third, it was used for LLM-assisted debugging—specifically in identifying the  $O(n^3)$  computational bottleneck of RBF kernels, which led to the selection of more scalable model architectures. Fourth, it assisted in optimizing the clarity of visualizations and the interpretability of evaluation outputs for stakeholder reporting.

At all times, the technical understanding, analytical reasoning, model selection decisions, and interpretation of results remained the responsibility of the team members. The AI served as a writing and debugging assistant rather than an analytical or experimental tool, and all findings in this report reflect the team’s own work on the dataset and the indoor localization problem.